INCH-POUND

MIL-S-13257H (AR)
AMENDMENT 7
6 February 2001
SUPERSEDING
AMENDMENT 6
29 March 1991

MILITARY SPECIFICATION

SIGNAL, ILLUMINATING, GROUND PARACHUTE, RED STAR, M126A1, WHITE STAR, M127A1 AND GREEN STAR, M195, PARTS AND LOADING, ASSEMBLING AND PACKING

This amendment forms a part of MIL-S-13257H (AR), dated 12 February 1980, and is approved for use by the U.S. Army Armament Research, Development and Engineering Center and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* 2.1:

Federal Standards, delete "FED-STD-151" in its entirety.

PAGE 2

2.1, Under Standards, add:

"COMMERCIAL
ASTM E8 - Tension Testing of Metallic Materials"

* 2.1, Drawings, delete "7548414" and "7548415" in their entirety and substitute the following:

"12900001 – Packing and Marking for Container, PA142 for Signal, Ground
12900009 – Packing and Marking for Container, M548, for Signals, Ground"

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* 3.3.1, Delete in its entirety and substitute the following:

"3.3.1 Premature burst. The assembly shall not burst within one hundred (100) feet from the launcher."

* Add new paragraphs 3.3.1.1 and 3.3.1.2 as follows:

"3.3.1.1 Premature ejection. The illuminant assembly shall not eject at an altitude of less than one hundred (100) feet from the ground within a horizontal distance of two hundred and fifty (250) feet from the launcher, or at an altitude of less than fifty (50) feet from the ground at a horizontal distance in excess of two hundred and fifty (250) feet from the launcher.

3.3.1.2 Launch delay. The signal must completely exit the rocket barrel within two (2) seconds of the striking of the primer."

* 3.3.3, Delete "the requirements of 3.3.1" and substitute "the requirements of 3.3.1, 3.3.1.1, and 3.3.1.2."

* 3.3.4a and b, Delete in its entirety and substitute the following:

"a. The requirements of 3.3.1, 3.3.1.1, 3.3.1.2, and 3.3.2.
b. In the M127A1 assembly (see note after 4.4.3.2.3.2)"

* 3.3.5, Delete in its entirety and substitute the following:

"3.3.5 Cold temperature. (-57 to -51°C or –70 to -60°F). The signal assemblies shall function and comply with the following requirements when tested as specified in 4.5.2.3.

a. The requirements of 3.3.1, 3.3.1.1, and 3.3.1.2.
b. The illuminant assembly burning time shall comply with 3.3.4b (See note after 4.4.3.2.3.2).
c. The average angle from the vertical of the ejected illuminant assemblies shall be not more than 25 degrees. The altitude shall be recorded for informational purposes.
d. The parachute shall not separate from the illuminant assembly. (Parachute opening delay shall be recorded for informational purposes as well as whether chute remains partially open)."

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* 3.4, Line 1, delete "metal container" and substitute "plastic container"

* 3.5, Delete in its entirety.

PAGE 4

* Add new paragraph 3.9 as follows:

"3.9 Airtightness. The metal container shall withstand, without leakage, an air pressure differential of three (3) pounds per square inch (psi) when tested as specified in 4.5.13."

PAGE 5

Table I, Under Signals, Assemblies, delete "Functioning, cold, hand fired ... 16 ... -- ... 3.3.5/4.5.2.3.2."

* Table I:

Delete "(Dwg. 7548414)" and substitute "(Dwg. 12900001)"

Under Containers (Dwg. 12900001), delete "Leakage test 96" and substitute "Leakage test 80"

Delete "Tear Strip Test (from the Leakage test)" in its entirety.

Add: "Container, Metal, M548
(Dwg. 12900009)
Examination for defects ... 4 metal containers 3/ ... 3.2 ... 4.4.2.24"

Add Note 3 as follows:

"3. If any of the four (4) M548 containers are found defective, it should be replaced with an acceptable container prior to functioning test."

PAGE 7

4.4.2.1, Tube casing, delete critical defects in their entirety and substitute the following:

"Critical				
1	Hydrostatic pressure test	100% (Note 1)	3.2	4.5.6
2	Pressure of cracks	100%	3.7	Visual

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Major				
101	Perpendicularity of ends	0.40%	3.2	Gage
102	Total length, max.	0.40%	3.2	Gage
103	Inside diameter at ends, max.	0.04%	3.2	Gage
Minor				
201	Poor workmanship	1.00%	3.8	Visual

Note 1 – Tubing stock used to manufacture the casing tube shall be 100% hydrostatically tested as specified in 4.5.6. All remaining classified defects shall be checked on the finished item."

PAGE 8

4.4.2.2, Bolt, Note a, delete "200 samples" and substitute "20 samples".

PAGE 10

4.4.2.4, Tail assembly, delete in its entirety and substitute the included page.

PAGE 11

4.4.2.5, Firing cap and spring clip assembly, Note a, delete "200 samples" and substitute "20 samples".

PAGE 25

4.4.2.19, Assembly, prior to assembling rocket assembly in rocket barrel:

Add: "Critical 2, Initiating charge missing … 100% … 3.2 … Visual"

Delete Major 103 in its entirety and substitute the following:

"103 Initiating charge over max (See note) … 0.40% … 3.2 … Balance."

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-S-13257H (AR)
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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	**Tail Assembly**				**8797947**
					NEXT HIGHER ASSEMBLY **8797968 or 9255112**
CLASSIFICATION	EXAMINATION/CHARACTERISTICS	NO. OF SAMPLE UNITS	AQL Or 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/INSPECTION METHOD
Critical:					
1	**Tail fin missing, cracked or inverted**		**100%**	**3.2**	**Visual**
Major					
101.	**Length, max.**		**0.40%**	**3.2**	**Gage**
102.	**Location of tail vanes with diameter and each other, including maximum vane extension from centerline**		**0.40%**	**3.2**	**Gage**
103.	**Inside diameter of upper tail ring, min**		**0.40%**	**3.2**	**Gage**
104.	**Perpendicularity of ends with diameter**		**0.40%**	**3.2**	**Gage**
105.	**Position of rib inside diameter with upper tail ring diameter**		**0.40%**	**3.2**	**Gage**
106.	**Weld missing**		**0.40%**	**3.2**	**Visual**
107.	**Weld flash excessive to the extent that function will be impaired**		**0.40%**	**3.2**	**Visual**
108.	**Weld peel test**	**(a)**	**--**	**3.2**	**4.5.11**
109.	**Mechanical properties (See dwg. 8797951)**	**(b)**	**--**	**3.2**	**4.5.8**
Minor					
201.	**Clearance between lowertail ring and tail vane**		**0.65%**	**3.2**	**Gage**
202.	**Poor workmanship**		**1.00%**	**3.8**	**Visual**

NOTES: **(a) 32 samples/lots, 1 upper tail ring weld and 1 lower tail ring weld per assembly (64 welds total); accept 0 reject 1. (b) Five (5) tail vanes/test samples shall be selected from each heat treated batch for mechanical properties testing.**

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* 4.4.2.19:

Add critical defect 3 as follows:

"3. Primer above flush … 100% … 3.2 … Gage"

Add Major defect 104 as follows:

"104 Depth of primer incorrect … 0.40% … 3.2 … Gage"

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* 4.4.2.23:

Delete Dwg. "7548414" and substitute "12900009"

Under next higher assembly, add "12900009"

Delete Critical 1 in its entirety.

Critical 2, add: "Cover" before "Embossing…"

Delete Major 103 and 106 in their entirety.

Add Major 107 as follows:

"107 Label missing or improperly applied … - … 0.40% … 3.2 … Visual"

Delete Minor 203 in its entirety.

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* 4.4.2.24, Delete in its entirety and substitute new paragraph as included page 30.

Add new paragraph 4.4.2.25 as included page 30a.

* Add new paragraph 4.4.2.26 as included page 30b.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.24	**Packing and Marking for Container, Metal M548 for Signals, Ground**				12900009
					NEXT HIGHER ASSEMBLY
CLASSIFICATION	EXAMINATION/CHARACTERISTICS	NO. OF SAMPLE UNITS	AQL Or 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/INSPECTION METHOD
Critical:	**None defined**				
Major					
101.	**PA142 Container quantity incorrect, or improperly inserted**		0.40%	3.2	Visual
102.	**Air tightness test (after sealing)**	1/		3.9	4.5.4/4.5.13
103.	**Metal container damaged (dents, splits, etc.)**		0.40%	3.2	Visual
Minor					
201.	**Contact loose**		0.65%	3.2	Visual
202.	**Marking misleading or unidentifiable**		0.65%	3.2	Visual
203.	**Metallic seal missing, unsealed, or improperly engaged**		0.65%	3.2	Visual
204.	**Poor Workmanship**		1.00%	3.8	Visual

NOTES:
1/ **One container out of each 20 containers packed out/sealed shall be tested, if the container fails the test, it should be removed from the lot and all containers back to the last successful test should be leak tested.**

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS **MIL-S-13257H (AR)**
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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.25	Signal Body				8797963
					NEXT HIGHER ASSEMBLY
CLASSIFICATION	EXAMINATION/CHARACTERISTICS	NO. OF SAMPLE UNITS	AQL Or 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/INSPECTION METHOD
Critical:	**None defined**				
Major					
101.	**Small inside diameter**		**0.40%**	**3.2**	**Gage**
Minor					
201.	**Large outside diameter, max.**		**0.65%**	**3.2**	**Gage**
202.	**Large inside diameter, min.**		**0.65%**	**3.2**	**Gage**
203.	**Overall length**		**0.65%**	**3.2**	**Gage**
204.	**Poor Workmanship**		**1.00%**	**3.8**	**Visual**

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF CHARACTERISTICS

MIL-S-13257H (AR)
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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.26	**Protector Assembly**				**8797941**
					NEXT HIGHER ASSEMBLY
CLASSIFICATION	EXAMINATION/CHARACTERISTICS	NO. OF SAMPLE UNITS	AQL Or 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE/INSPECTION METHOD
Critical:	**None defined**				
Major					
101.	**Outside diameter, max.**		**100%**	**3.2**	**Gage**
Minor	**None defined**				
NOTES:					

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4.4.3.1, Last sentence of note, delete "4.5.2" and substitute "4.5.1"

PAGE 32

4.4.3.2, Table III, add the following inspection:

"Cork sealing disc in end of barrel loose (See label dwgs. 8797931-2, 3 and 10) ... Major"

* Table III:

Add the following characteristics:

"Premature ejection	(See 3.3.1.1)	Special (see 6.8)
Launch delay	(See 3.3.1.2)	Major
Parachute separation at cold temperature		Major"

Delete "Assembly bursts at a distance of more than one hundred (100) feet from the launcher (See 3.3.1) Major" and substitute "Assembly bursts at a distance of more than one hundred (100) feet from the launcher but less than the distance specified in 3.3.1.1 Special"

4.4.3.2.1.1, Delete "one hundred (100) assemblies" and substitute "fifty (50) assemblies"

4.4.3.2.1.2, Delete "twenty-five (25) assemblies" and substitute "twenty (20) assemblies"

* 4.4.3.2.1.1 and 4.4.3.2.1.2, Delete "a critical defect" and substitute "a critical or special defect"

4.4.3.2.2.1:

Delete "one hundred and twenty five (125) assemblies" and substitute "one hundred (100) assemblies"

Delete "or if eight (8) or more" and substitute "or if six (6) or more"

* 4.4.3.2.2.1, Delete "a critical defect occurs" and substitute "a critical or special defect occurs"

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* 4.4.3.2.2.2, Delete "a critical defect occurs" and substitute "a critical or special defect occurs"

4.4.3.2.3.1, Delete in its entirety and substitute the following:

4.4.3.2.3.1 Beginning with the first lot produced and continuing until three (3) consecutive lots have been found to be acceptable, fifty (50) assemblies shall be selected from each lot for test. The lot shall be rejected if, during the test a critical defect occurs, the average angle from the vertical exceeds the requirements, or if four (4) or more assemblies exhibit any of the remaining major defects listed in Table III. (See note after the following paragraph)."

4.4.3.2.3.2, Delete in its entirety and substitute the following:

"4.4.3.2.3.2 After three (3) consecutive lots have complied with the criteria of 4.4.3.2.3.1, thirty-two (32) assemblies shall be selected from each lot for test. The lot shall be rejected, if during the test, a critical defect occurs, the average angle from the vertical exceeds the requirement or if three (3) or more assemblies exhibit any of the remaining major defects listed in Table III. (See note).

Note: A star which ignites and burns normally but subsequently fails to meet minimum burning time requirements due to ground impact, will not be classified defective. Firing records will be annotated accordingly."

* 4.4.3.2.3.1 and 4.4.3.2.3.2, Delete "a critical defect occurs" and substitute "a critical or special defect occurs"

4.4.3.2.4, Delete in its entirety.

PAGE 34

4.4.3.3, Second sentence, delete "as classified in Table III of 4.4.3.2" and substitute "(See 3.3.1)"

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4.5.2.1:

* Line 4, delete " metal" and substitute "plastic"

Note, delete "60 seconds" and substitute "three minutes"

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4.5.2.2, Note, delete "60 seconds" and substitute "three minutes"

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4.5.2.3.1:

* Line 4, delete "metal" and substitute "plastic"

Note, delete "60 seconds" and substitute "three minutes"

4.5.2.3.2, Delete in its entirety.

* 4.5.3:

Delete "Leakage" and substitute "Leakage (plastic container)"

Line 1, delete "The packed sealed container" and substitute "The packed plastic container (dwg. 12900001)"

* 4.5.4, Delete in its entirety.

PAGE 38

4.5.6, Delete in its entirety and substitute the following:

"4.5.6 Hydrostatic test of tube, casing. In lieu of testing each individual casing tube, the tubing stock, which is used to manufacture the casing tube, shall be 100% tested in a hydrostatic test device (see 4.4.4). The minimum pressure shall be applied for not less than 5 seconds. The tubing shall be observed for evidence of failure to comply with the requirements. Any full length of tubing that bursts, cracks, or shows evidence of distortion shall be classified as defective and removed from production."

4.5.8, Delete in its entirety and substitute the following:

"4.5.8 Mechanical properties of tail vane. The tail vane/test samples shall be tested for mechanical properties in accordance with ASTM E8. If any tail vane/test sample fails to comply with the criteria specified on the applicable drawing, the heat treated batch represented by the sample shall be rejected."

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4.5.10, Delete the last sentence and substitute the following:

"If one or more assemblies fail to comply with the color value requirement or if two or more assemblies fail to meet the candlepower requirement, the lot shall be rejected."

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* Add the following paragraphs:

"4.5.13 Airtightness. The metal container shall be conditioned to ambient temperature and pressure prior to testing. The container shall be tested in accordance with 4.5.13.1 or with permission of the procurement agency with 4.5.13.2. A wetting agent may be used to minimize air bubbles clinging to the exterior surface. A stream or recurring succession of bubbles from any surface, seam or junction shall be evidence for a defective container. Any container that fails to comply with the requirement shall be classified defective and removed from the lot.

4.5.13.1 Vacuum method. The container shall be tested for leakage by immersing the closed container in the inverted position under water in a vacuum vessel and lower the pressure in the vessel to three pounds per square inch minimum below ambient pressure. Observation for leakage of air from the container interior shall be made for a minimum of three seconds after reduction of pressure.

4.5.13.2 Hot water method. The container shall be tested for leakage by immersing the closed container in the inverted position to a depth of one inch below the surface of the water. The temperature of the water and the length of time of immersion shall be that which will assure an increase in pressure to a minimum of 3 psi above ambient pressure. Observation for leakage of air from the container interior shall be made during the period of pressure buildup and for 30 seconds after the 3 psi pressure differential has been reached."

* 5.1.1, Delete "7548414" and substitute "12900001"

* 5.2.1, Delete "7548415" and substitute "12900009"

* 5.3, Delete "7548414 and 7548415" and substitute "12900001 and 12900009"

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6.4, Delete in its entirety.

6.7, Delete the table and substitute the following:

"PRODUCTION LOTS

Firing Method & Conditioning Temp	First Article	First three Consecutive Acceptable	Normal Production	Requirement Paragraph
Fixture:				
68° to 74°C (155° to 165°F)	32	50	20	See 3.3.3
18° to 24°C (65° to 75°F)	32	100	50	See 3.3.4
-57° to -51°C (-70° to -60°F)"	16	50	32	See 3.3.5

* Add 6.8 as follows:

"6.8 Definitions.

a. Special defect. A defect, other than critical that judgment and experience indicates may, depending upon the degree of variance from the design requirement:

1. Results in hazardous or unsafe condition for individuals using, maintaining or depending upon the product.
2. Prevent performance of the tactical function of a major end item.

b. Premature burst. A signal which ejects, ignites and burns at a distance less than that specified in the requirement paragraph (see 3.3.1).

c. Premature ejection. A signal in which the illuminant assembly ejects but fails to ignite (see 3.3.1.1)."

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AMENDMENT 7

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